

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. How has “The Toyota Way” been the roots of DevOps? Explain.
 - b. What are the three transformational forces responsible of the wicked business problems? Discuss.
 - c. What are the eight elements of waste? Give examples and business outcomes of each.
 - d. As DevOps is not a formal framework, it leads to suboptimal practices. What are they?

2. **Attempt any two of the following:** 12
 - a. What are the advantages of API strategy? Explain.
 - b. What are the new skills required by QA and testing professionals to become better skilled?
 - c. Discuss the five levels of continuous delivery maturity.
 - d. What is shift-left monitoring? Explain the continuous high-quality feedback and Intelligence and Analytics.

3. **Attempt any two of the following:** 12
 - a. Discuss the issues and challenges in DevOps adaptation.
 - b. Explain the business canvas model and its components.
 - c. What are Key Partnerships and Cost Structures in the business canvas model?
 - d. Discuss the Business Intent: Optimisation versus Innovation.

4. **Attempt any two of the following:** 12
 - a. What are the new fluid guidelines and principles for enterprise architecture? Explain each.
 - b. State and explain the seven factors that lead to normalization of deviance.
 - c. What are the wasteful practices traditionally associated with manufacturing that are applicable to government software development context? Explain.
 - d. What is Uber syndrome? What is the role of technology in innovation? Explain.

5. **Attempt any two of the following:** 12
 - a. What are the capabilities and goals of DevOps Center of Competency? Discuss.
 - b. How can we develop the culture of continuous improvement? Explain.
 - c. What are the anti-patterns for DevOps adoption?
 - d. How does one think for Line of Business in the DevOps way?